DATA 8014 — Assignment 1
Exploring Dictionary Learning

TODO: Name and student ID

Fall 2026

Repository. **TODO: Insert the URL of the private repository in the `hku-data8014-students` organization.**

1 Implementation and experimental protocol

I added a NumPy/SciPy package, `data8014/dictlearn`, containing the synthetic generators, recovery and validation metrics, ISTA/FISTA, MSP, alternating minimization, convergence fitting, and MNIST PCA/whitening. Thin experiment drivers in `scripts/dl_part*.py` write incremental JSONL logs and regenerate figures from those logs. Part 3 uses `float32`; the synthetic experiments use `float64`. Unit tests check moments and dictionary constraints, sign/permutation invariance, the exact one-step ISTA solution for an orthogonal dictionary, step-size calculations, and the Part 4 variance construction.

Unless otherwise stated, reported uncertainty is the sample standard deviation across independent seeds. I used five seeds in Parts 0–2 and 4 and three seeds in the more expensive MNIST experiments. For recovery I primarily use (3), report duplicate matches, and additionally log a Hungarian one-to-one score. Thus a collapsed dictionary cannot silently appear as successful. Validation codes are always inferred from held-out samples. In the complete case, because every MSP iterate is orthogonal, ISTA from zero gives the exact LASSO solution in one step,

$$\widehat{Z} = \mathrm{soft}(X_{\mathrm{val}}\widehat{A}^\top, \lambda/2).$$

My MSP update is

$$A_{t+1} = \mathcal{P}_{O(D)}\Big(((XA_t^\top)^{\circ 3})^\top X\Big),$$

where the polar projection is computed by an SVD. One alternating-minimization block consists of 30 warm-started ISTA steps, one dictionary-gradient step, and row normalization. I recompute $\eta = (2\|A\|_2^2)^{-1}$ and $\nu = (2\|Z\|_2^2)^{-1}$ in every block.

2 Part 0: setup and warm-ups

2.1 Generator, metrics, and MSP sanity check

For the Bernoulli–Gaussian generator I draw $z_k = b_k g_k$, $b_k \sim \mathrm{Bern}(\theta)$ and $g_k \sim \mathcal{N}(0, 1/\theta)$. Complete dictionaries are QR factors; overcomplete dictionaries have independently drawn, unit-normalized rows.

With the prescribed MSP sanity setting, all five seeds recovered every atom to $s_k > 0.99$ in 6–8 one-based iterations. Final minimum correlations ranged from 0.9986 to 0.9993.

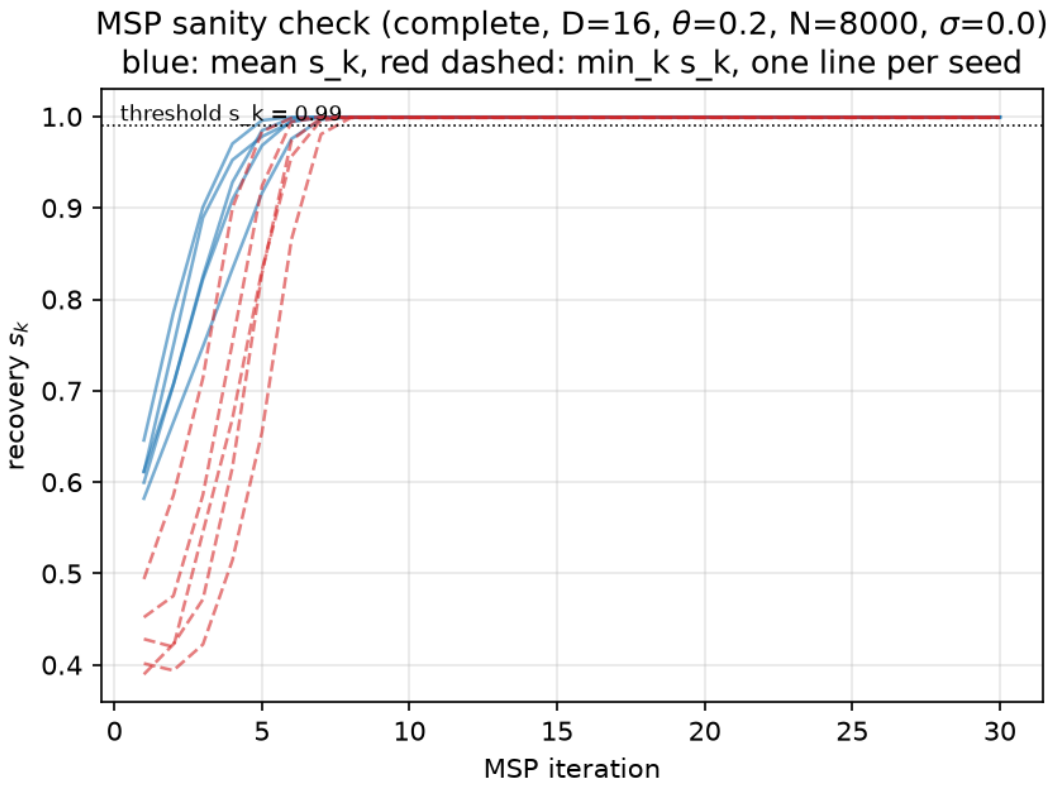


Figure 1: MSP sanity check. The minimum atom correlation exceeds 0.99 well before the 30-iteration reference budget for every seed.

2.2 ISTA convergence

For the fixed ground-truth overcomplete dictionary, the smooth term has Lipschitz constant $L = 2\|A\|_2^2$. The plotted worst-case proximal-gradient bound is

$$f(Z_k) - f^\star \le \frac{L\|Z_0 - Z^\star\|_F^2}{2k}.$$

I estimated $f^\star = 8814.0855168404$ by a long FISTA run. The solution has 8273 nonzero entries out of 64000. ISTA permanently identifies this entrywise support at iteration 2273; after that, the fitted geometric factor is $\rho = 0.99783$ per iteration. The eventual geometric behavior is faster than the generic $O(1/k)$ guarantee. The bound does not use the finite active-set structure: after signs and support stabilize, soft thresholding is affine and the remaining iteration is a linear contraction on the active coordinates.

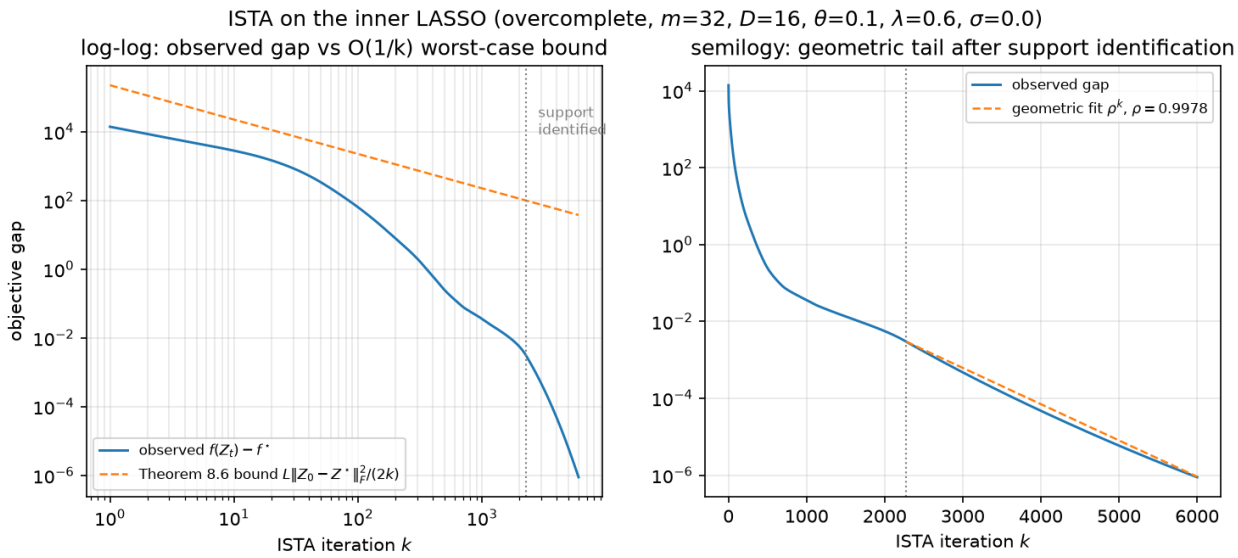


Figure 2: ISTA objective gap against the worst-case convex bound (left) and the geometric tail after support identification (right).

2.3 Why orthogonality costs no generality

Let a noiseless complete model have an arbitrary invertible A and $x = A^\top z$. Since the coordinates of z have unit variance,

$$\Sigma_x = \mathbb{E}[xx^\top] = A^\top A.$$

With the symmetric whitening matrix $W = \Sigma_x^{-1/2}$, transform row samples as $\bar{X} = XW$ and define $\bar{A} = AW$. Then

$$\bar{A}^\top \bar{A} = W^\top A^\top A W = I.$$

Because $\bar{A}$ is square, it is orthogonal. Learning the whitened model $\bar{X} = Z\bar{A}$ therefore loses no information: the original dictionary is $A = \bar{A}W^{-1}$. In finite samples I use the empirical covariance; this argument requires the stipulated invertibility and zero-noise setting.

3 Part 1: convergence and sample-size scaling

3.1 Optimization convergence (T-axis)

I used $D = 16$, $N = 8000$, $\theta = 0.1$, $\sigma = 0$, $\lambda = 0.6$, and five seeds. Both metrics reach their finite-sample/LASSO plateaus near $T = 12$. To avoid fitting the plateau as if it were decay, I estimated a tail floor and fit only the positive transient above it. Table 1 compares the mean-curve fits; the per-seed parameter spread is also shown.

Table 1: Part 1(a) floor-aware transient fits. Larger R^2 is better.

Metric	floor	power α	R^2	exp. ρ	R^2	per-seed ρ
Recovery error	3.61×10^{-4}	4.307	.592	.350	.799	$.210 \pm .126$
Eval loss	.1470	4.957	.608	.324	.814	$.257 \pm .098$

The exponential model is more consistent with both measured transients, but $R^2 \approx 0.8$ and the short, very rapid approach to a floor mean that neither simple law is a complete global description. Recovery tends to zero up to finite-sample error, whereas the eval loss retains a nonzero LASSO shrinkage floor, so the raw metrics do not share the same asymptote.

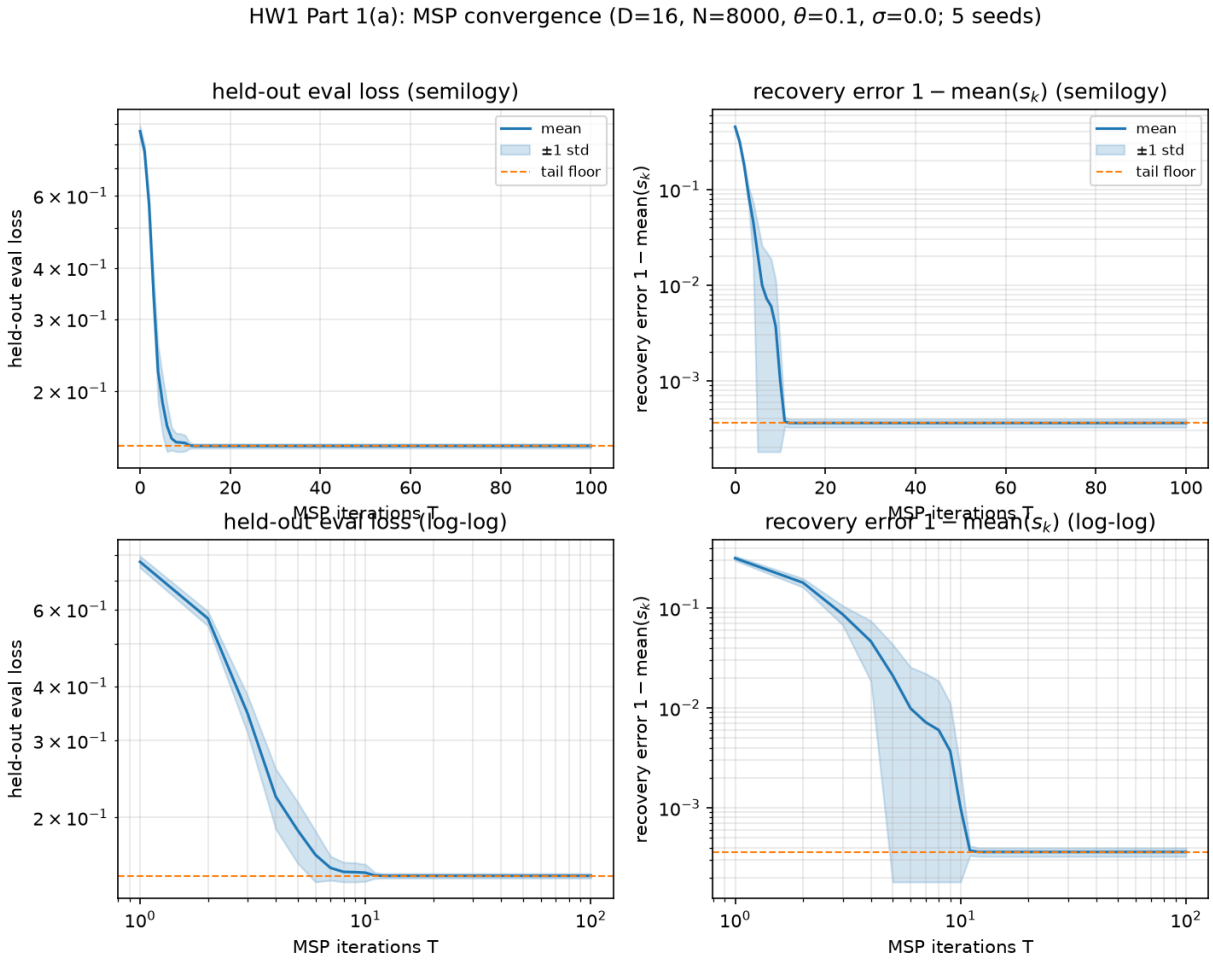


Figure 3: Part 1(a): held-out and recovery errors on semilogy and log–log axes. Bands show one standard deviation over five seeds.

3.2 Sample-size scaling (N-axis)

For each seed I drew the $N = 10000$ dataset once and used prefixes, holding the true dictionary, initialization, and validation data fixed. I stopped using a signed/permuted iterate-stationarity criterion rather than ground-truth recovery. All 45 runs converged before the cap (maximum 126 iterations). The first six of nine N values were fit and $N = 10000$ was predicted.

Table 2: Part 1(b) fits of $\mathcal{E} - \mathcal{E}_\infty = cN^{-\beta}$.

Metric	mean-curve β	R^2	per-seed β	target relative error
Recovery error	1.285	.936	$1.212 \pm .192$	16.1%
Eval excess	.472	.907	$.463 \pm .077$	207%

The raw eval-loss prediction error is only 13.9%, but this is misleading because the estimated floor $\mathcal{E}_\infty = 0.13673$ dominates. Relative to the quantity actually fit, $\mathcal{E} - \mathcal{E}_\infty$, the extrapolation error is 207%. Thus recovery is reasonably compatible with a power law over the measured range, while eval-excess extrapolation is not stable enough to support the same claim despite its in-range R^2.

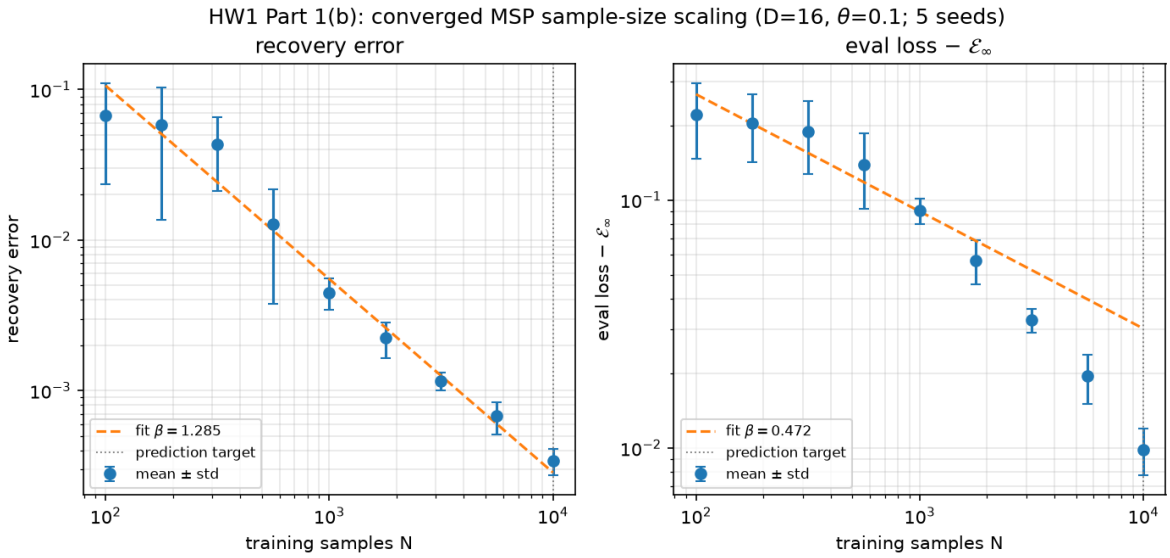


Figure 4: Part 1(b): converged recovery error and held-out excess loss versus sample count.

4 Part 2: density and identifiability

4.1 Cost and transition

For $\theta \in \{.2, .4, .6, .7, .8, .85\}$, the mean $T_\star$ values were 8.0, 9.8, 17.8, 26.6, 48.0, and 88.0; their standard deviations were 2.55, 1.79, 1.30, 8.17, 4.85, and 16.51, respectively. No run was censored. After recording the first-passage time, I continued each run to iterate stationarity before evaluating its held-out loss. Density therefore causes a large and increasingly variable optimization cost.

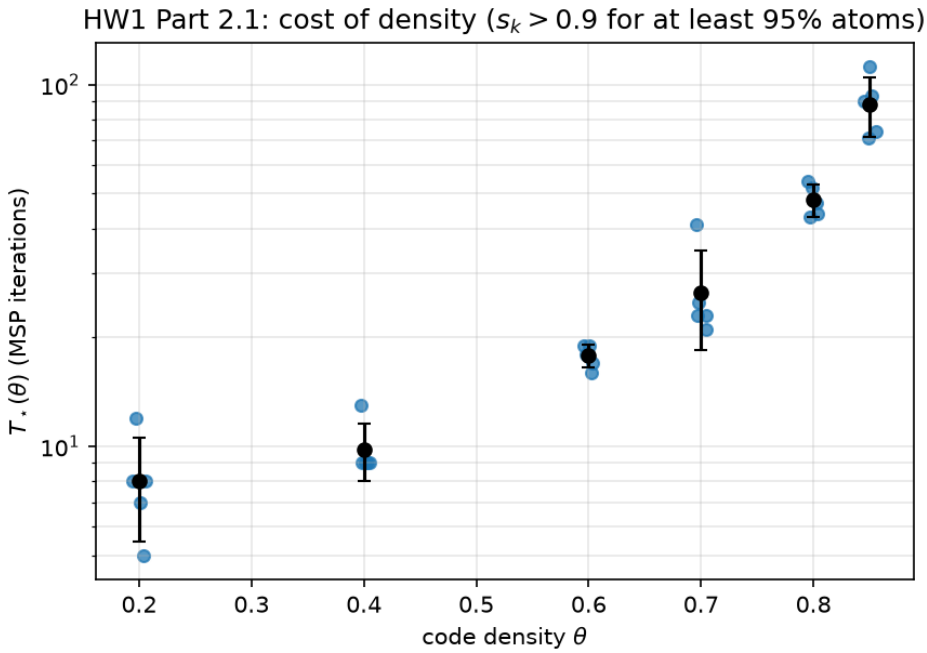


Figure 5: Iterations required to recover at least 95% of atoms above 0.9.

At higher density the mean recovered fractions were

θ	.88	.90	.91	.92	.93	.95	1.00
fraction	.9875	.9375	.4875	.175	.075	0	0
std.	.028	.088	.277	.103	.052	0	0

All runs reached iterate stationarity before the 20000-iteration cap. The sharp fall between .90 and .92 is evidence of a finite-sample/algorithmic phase transition. In contrast, converged held-out loss rises smoothly: its mean is .2815 at $\theta = .2$, 1.1709 at .90, 1.1935 at .91, 1.2043 at .92, and 1.2118 at 1. There is no loss discontinuity corresponding to the recovery collapse, so reconstruction is not an identifiability diagnostic.

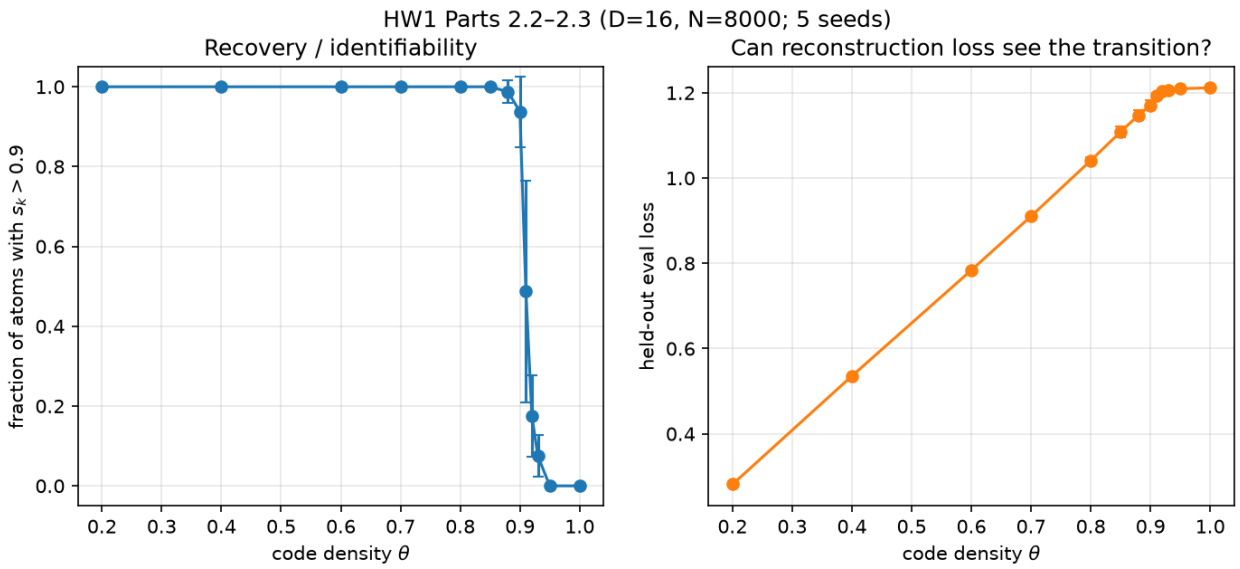


Figure 6: Part 2 transition. Recovery changes sharply while held-out reconstruction loss changes smoothly.

4.2 Why the transition occurs

For $z = bg$, $b \sim \text{Bern}(\theta)$ and $g \sim \mathcal{N}(0, 1/\theta)$,

$$\mathbb{E}[z^2] = \theta \frac{1}{\theta} = 1, \qquad \mathbb{E}[z^4] = \theta\, 3 \left(\frac{1}{\theta}\right)^2 = \frac{3}{\theta}, \qquad \kappa(\theta) = 3\left(\frac{1}{\theta} - 1\right).$$

At the observed transition center $\theta = .91$, $\kappa \approx .297$; at $\theta = 1$, $\kappa = 0$.

At $\theta = 1$, $z \sim \mathcal{N}(0, I)$. For every orthogonal R, set $Z' = ZR^\top$ and $A' = RA$. Then $Z'A' = ZA$, while rotational invariance gives $Z' \overset{d}{=} Z$. Infinitely many dictionaries therefore induce the same data law. MSP's fourth-power objective exploits coordinate-wise non-Gaussian kurtosis: sparse directions produce unusually large fourth moments when the iterate aligns with the independent components. As $\theta \to 1$, the excess kurtosis and hence this directional contrast vanish. The smooth loss remains small because a rotated Gaussian representation reconstructs just as well.

5 Part 3: MNIST

5.1 Spectrum, projection, and whitening

I centered the MNIST training images, diagonalized their pixel covariance, and used $k = 32$. The 80%, 90%, and 99% variance thresholds require 44, 87, and 331 components, respectively, so $k = 32$ deliberately trades retained variance for computational cost. I divided each projected coordinate by the square root of its training eigenvalue. The resulting relative covariance error was $\|\widehat{\Sigma}_y - I\|_F / \|I\|_F = 2.77 \times 10^{-7}$.

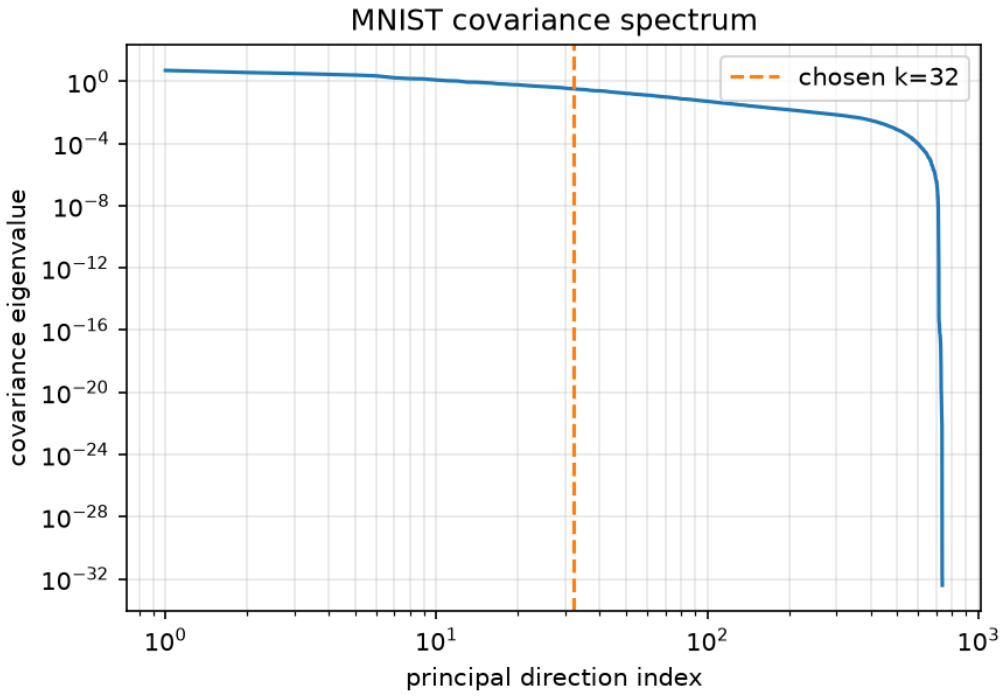


Figure 7: MNIST covariance spectrum and cumulative explained variance.

5.2 T-axis

I trained on $N = 12000$ images for 400 blocks with $\lambda = 2$ and evaluated on a fixed, unused set of 2000 test images. Table 3 reports the final mean and standard deviation.

Table 3: Part 3 T-axis results after 400 blocks (three seeds).

m	held-out loss	random loss	learned/random	code nnz fraction
64	$10.1889 \pm .0098$	$15.9189 \pm .3085$	$.6402 \pm .0128$	$.13050 \pm .00084$
128	$7.9752 \pm .0238$	$11.8963 \pm .0640$	$.6704 \pm .0043$	$.07403 \pm .00040$

Floor-aware held-out fits favored exponential decay: for $m = 64$, $\rho = .9789$ with $R^2 = .954$ versus power $R^2 = .841$; for $m = 128$, $\rho = .9850$ with $R^2 = .822$ versus power $R^2 = .774$. Training-objective exponential fits had $R^2 = .938$ and .941. Thus the real-data overcomplete problem also has an approximately geometric transient, but it relaxes over hundreds of blocks rather than MSP's roughly twelve iterations, and the late curves are dominated by a floor and small nonmonotonic fluctuations.

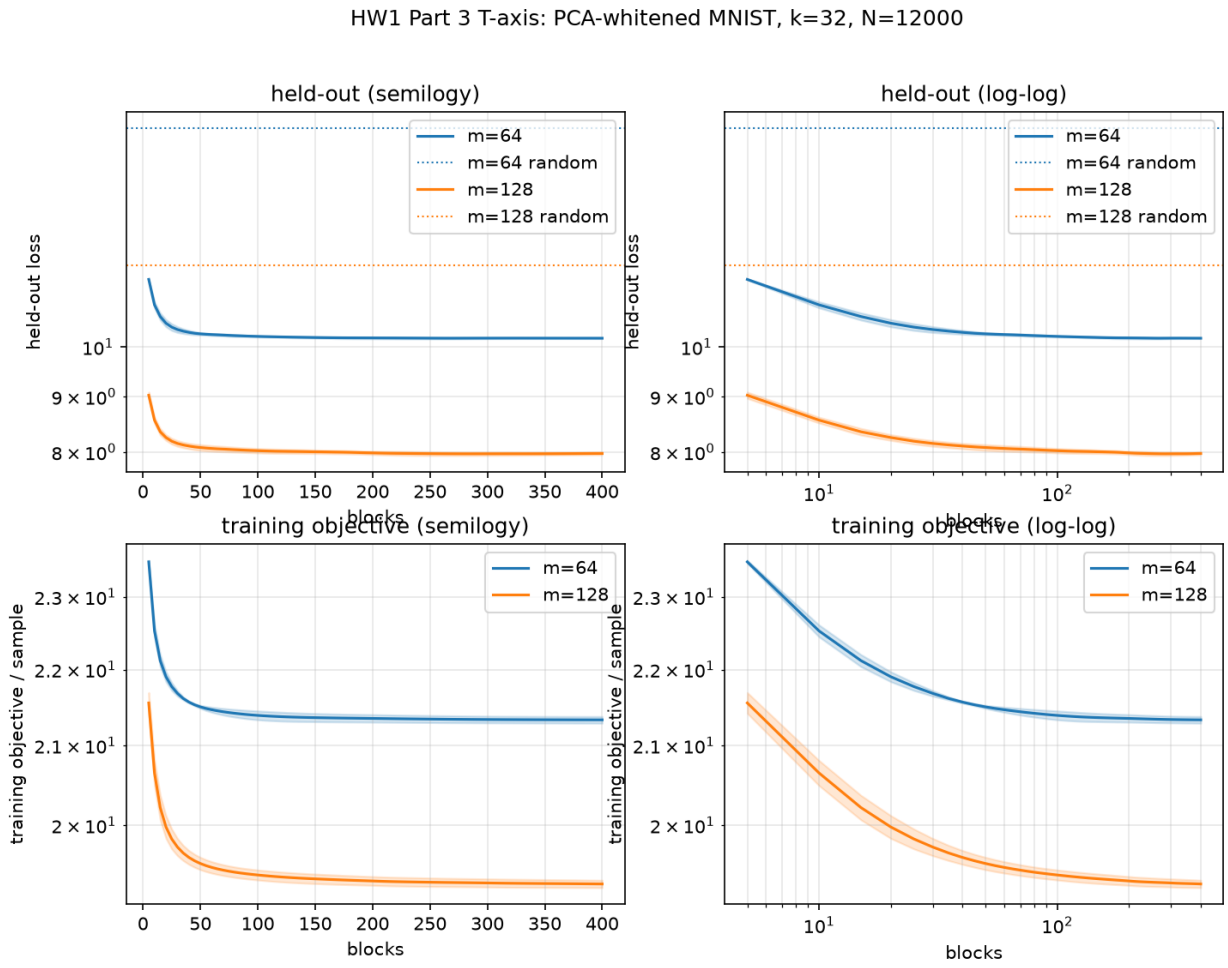


Figure 8: MNIST T-axis, with one-standard-deviation bands and random-dictionary held-out baselines.

5.3 N-axis

For each seed I fixed an image ordering and used prefixes at the seven required sample counts. I fit the smallest five values with a free asymptotic floor and predicted $N = 12800$.

Table 4: MNIST sample-size fits and diagnostics.

m	β	per-seed β	R^2	prediction error	capped runs
64	.558	$.555 \pm .155$	.9999	1.22%	3/21
128	.485	$.494 \pm .184$	.9996	1.54%	12/21

The large R^2 and small target errors support a floor-plus-power description on these points, but 15 of 42 runs reached the permitted 1000-block cap rather than the 10^{-5} stopping criterion. This is especially important for $m = 128$ and limits precision. The synthetic eval exponent was $.463 \pm .077$ per seed. The $m = 128$ MNIST exponent is nearly identical and $m = 64$ is modestly steeper, but the MNIST across-seed standard deviations are large enough that this difference is not decisive. At $N = 12800$, the learned/random ratios were $.6408 \pm .0129$ and $.6711 \pm .0033$, and code nnz fractions were $.13063 \pm .00025$ and $.07421 \pm .00014$, confirming learning relative to the random baseline.

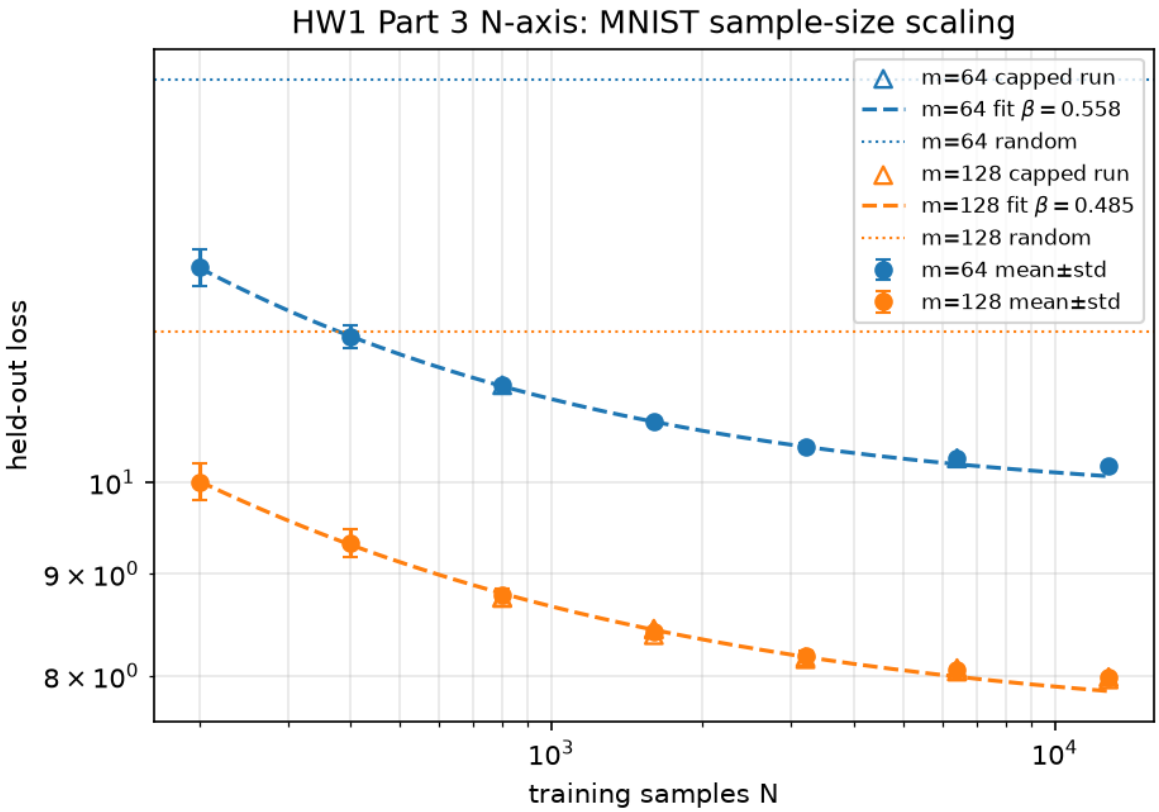


Figure 9: MNIST N-axis fits; triangular markers identify capped runs.

5.4 Atoms

To visualize an atom a in whitened PCA coordinates, I applied the inverse linear transform

$$a_{\text{pixel}} = a \, \text{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_k}) V_k^\top.$$

I did not add the data mean to a direction. Both dictionaries contain recognizable digit and stroke templates. The $m = 128$ dictionary contains more writing-style variants and near-duplicate parts. Greater overcompleteness lowers absolute held-out loss and makes codes sparser, while individual atoms remain interpretable.

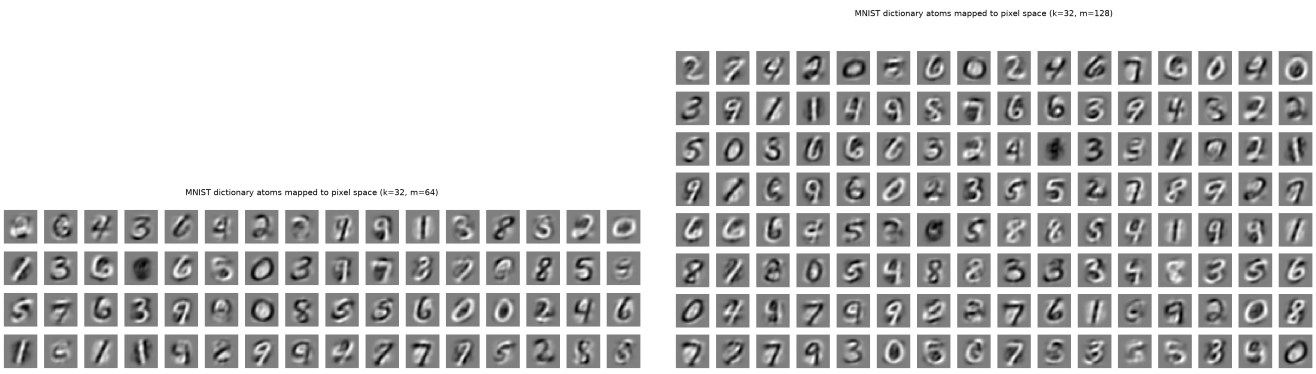


Figure 10: Inverse-whitened MNIST atoms for $m = 64$ (left) and $m = 128$ (right).

6 Part 4 (bonus): spectral bias

6.1 Derivation and generator check

Since $\mathbb{E}[b_k g_k] = 0$ and coordinates are independent,

$$\mathbb{E}[z] = 0, \qquad \mathbb{E}[zz^\top] = \operatorname{diag}(p_1 v_1, \ldots, p_m v_m) = \operatorname{diag}(\tau_1, \ldots, \tau_m).$$

Consequently, for $x = A^\top z + \epsilon$,

$$\operatorname{Cov}(x) = A^\top \operatorname{diag}(\tau_k) A + \sigma^2 I.$$

If the rows $a_k^\top$ of square A are orthonormal, then $\operatorname{Cov}(x) a_k = (\tau_k + \sigma^2) a_k$: each atom is an eigenvector with eigenvalue $\tau_k + \sigma^2$.

Here $p_k = ck^{-\alpha}$ and $v_k = 1$, so expected sparsity is $\sum_{k=1}^m p_k = c \sum_{k=1}^m k^{-\alpha}$, which decreases with α (for all $k > 1$). For $m = 16, c = .45$, its values are 2.079, 1.521, and 1.175 at $\alpha = .75, 1, 1.25$. If instead $v_k = 1/p_k$, then $\tau_k = 1$ and every signal eigenvalue is equal. This explains why the variance-normalized Part 0 generator cannot test spectral bias. The empirical coordinate variances in Figure 11 track the intended unequal τ_k.

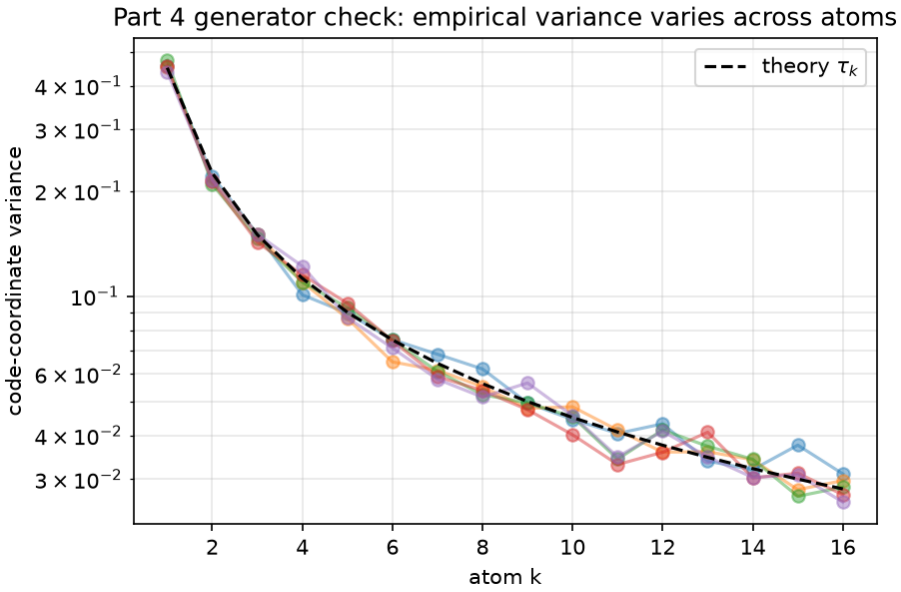


Figure 11: Empirical code variances for five seeds at $\alpha = 1$.

6.2 Recovery ordering and algorithm dependence

With alternating minimization, high-energy atoms are generally recovered first and the separation strengthens as α increases. Seed variation is visible but smaller than the overall high-to-low-energy trend. Unrecovered atoms are shown above the budget line.

Table 5: Part 4 recovery summaries. Mean time is first averaged within each seed over recovered atoms; ranges pool recovered atom–seed pairs.

Method	α	recovered fraction	mean recovery time	pooled range
Alt. min.	.75	$1.000 \pm .000$	17.28 ± 5.26	2–58
Alt. min.	1.00	$.975 \pm .056$	24.11 ± 7.37	1–79
Alt. min.	1.25	$.900 \pm .084$	32.95 ± 11.69	1–113
MSP	1.00	$1.000 \pm .000$	$2.83 \pm .29$	1–6

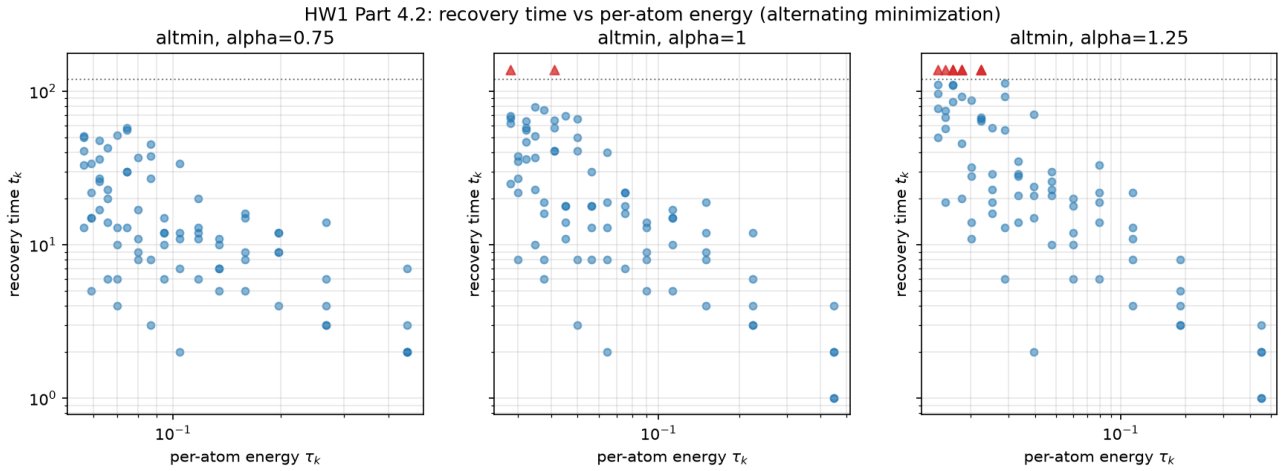


Figure 12: Alternating-minimization recovery times. Red triangles are atoms not recovered within 120 blocks.

At $\alpha = 1$, MSP recovers all 80 atom–seed pairs within six iterations and shows little persistent ordering by energy. Unlike coordinate inference and a gradual dictionary gradient step in alternating minimization, MSP applies a fourth-moment stretch followed by a polar projection, updating the entire orthogonal basis jointly. This global coupled projection rapidly enters its local recovery phase and largely removes the slow, coordinate-by-coordinate spectral ordering.

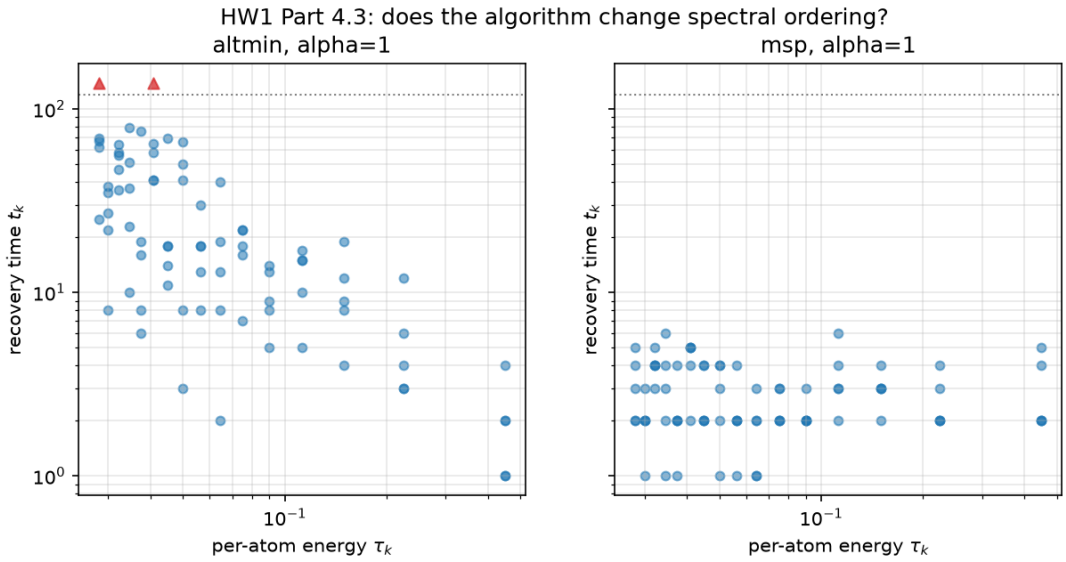


Figure 13: Identical $\alpha = 1$ data and initializations: alternating minimization (left) versus MSP (right).

7 Reproducibility and limitations

All required dimensions satisfy $D \leq 64$, $m \leq 256$, and $N \leq 20000$. Random seeds, configurations, stopping states, censored/capped flags, and per-seed measurements are stored in JSONL logs. The main limitations are the short floor-limited transient in Part 1(a), unstable eval-excess extrapolation in Part 1(b), finite-sample and algorithm dependence of the observed Part 2 transition, and 15 capped MNIST N-axis runs. These caveats are retained rather than silently filtering runs.

8 AI usage disclosure

I used an OpenAI coding assistant through the Pi coding-agent interface to help inspect the assignment, audit the experimental plan, implement and review NumPy experiment code, diagnose the local Python/LAPACK environment, run the experiments, check plots and logs, and prepare a draft of this report. The assistant also helped identify and correct issues including the sign of the Bernoulli–Gaussian excess kurtosis, the orthogonal reparameterization at $\theta = 1$, inverse-whitening of atoms, stationarity-based stopping, and continuing Part 2 runs beyond the first-passage time before evaluating loss. All numerical claims in this document come from the repository's saved logs and reproducible scripts. **TODO: Before submission, personally verify the code, figures, derivations, wording, repository URL, and that this disclosure accurately describes your use and understanding.**